Central Limit Theorem

Statistics II

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Additivity (Normal Distribution)

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If:

1. $X_i \sim Normal(\mu, \sigma), i = 1, \ldots, n$ — **independent and identically distributed random variables**
2. $T = X_1 + \ldots + X_n$ and $\bar{X} = \dfrac{T}{n}$

then:

$$T \sim Normal(\mu_T = n \times \mu, \sigma_T = \sqrt{n} \times \sigma)$$

$$\bar{X} \sim Normal\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$$

Example 1

Consider:

1. Independent variables: $X_i \sim Normal(\mu_i = 40, \sigma_i = 10)$

   X_i — lunch duration of student i (in minutes)

2. $\bar{X} = \dfrac{1}{n}(X_1 + \ldots + X_n)$ — sample mean ($n = 25$ and $n = 81$ students)

Find the probability that the sample mean lies between 39 and 41 minutes.

Solution: (assuming Normal distribution)

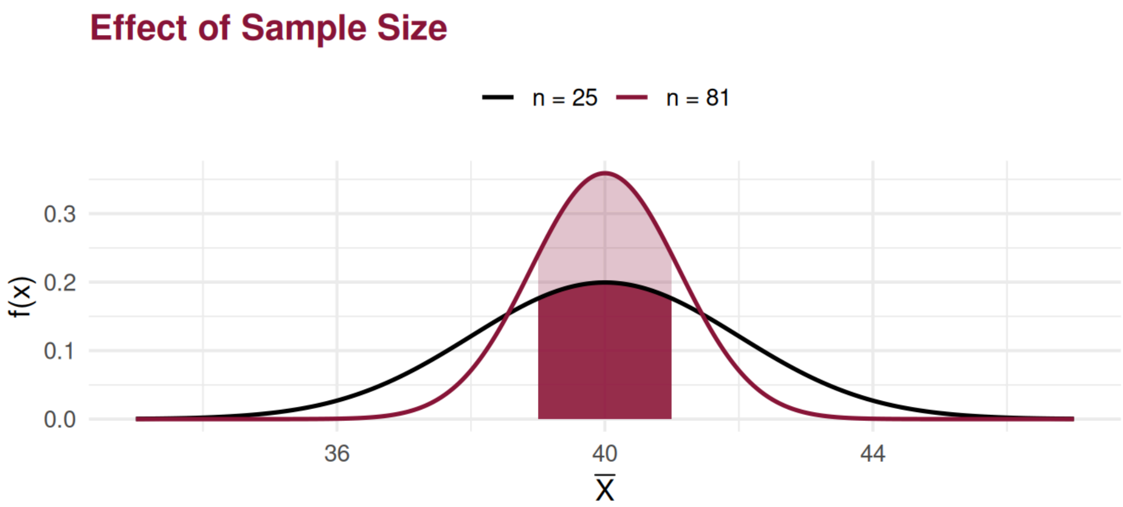
$$\bar{X} \sim Normal\left(\mu_{\bar{X}} = 40, \sigma_{\bar{X}} = \frac{10}{\sqrt{n}}\right)$$

$$\Rightarrow P(39 < \bar{X} \leq 41) = P\left(\frac{39-40}{\frac{10}{\sqrt{n}}} < Z \leq \frac{41-40}{\frac{10}{\sqrt{n}}}\right)$$

Example 1 — Solution (cont.)

$P(39 < \bar{X} \leq 41) =$

- $n = 25$: $= P(-0.5 < Z \leq 0.5) = \Phi(0.5) - \Phi(-0.5) = \mathbf{0.3829}$
- $n = 81$: $= P(-0.9 < Z \leq 0.9) = \Phi(0.9) - \Phi(-0.9) = \mathbf{0.6319}$



> **Comment:** As sample size increases, the probability that the sample observations concentrate around the mean also increases.

Central Limit Theorem (*CLT*)

Central Limit Theorem (*CLT*):

If:

1. $X_i \sim Distribution(\mu, \sigma)$, $i = 1, \ldots, n$ — **independent and identically distributed random variables**
2. $T = X_1 + \ldots + X_n$, $\quad \bar{X} = \dfrac{T}{n}$

then, for n sufficiently large (as a rule, $n \geq 30$):

$$T \dot{\sim} Normal(\mu_T = n \times \mu, \sigma_T = \sqrt{n} \times \sigma)$$

$$\bar{X} \dot{\sim} Normal\left(\mu_{\bar{X}} = \mu, \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\right)$$

Example 1 — Revisited (CLT)

Consider:

1. Independent variables: $X_i \sim Distribution(\mu_i = 40, \sigma_i = 10)$

   X_i — lunch duration of student i (in minutes)

2. $\bar{X} = \frac{1}{n}(X_1 + \ldots + X_n)$ — sample mean

   $n = 25,\ n = 81,\ n = 225$ students

Find $P(39 < \bar{X} \leq 41)$. Solution: (without the assumption of normality)

With $\mathbf{n \geq 30}$:

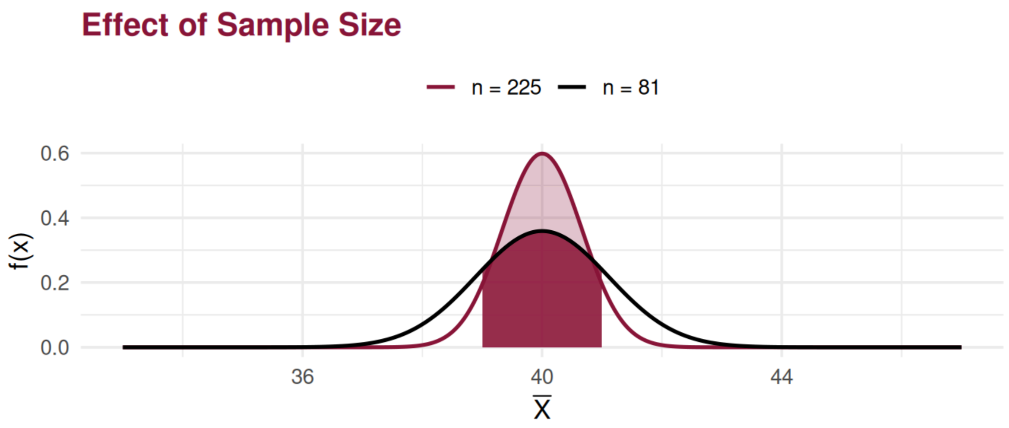
$$\bar{X} \stackrel{.}{\sim} Normal\left(\mu_{\bar{X}} = 40, \sigma_{\bar{X}} = \frac{10}{\sqrt{n}}\right) \xrightarrow{CLT}$$

$$\Rightarrow P(39 < \bar{X} \leq 41) \approx P\left(\frac{39-40}{\frac{10}{\sqrt{n}}} < Z \leq \frac{41-40}{\frac{10}{\sqrt{n}}}\right) =$$

Example 1 — Revisited (cont.)

$P(39 < \bar{X} \leq 41) \rightarrow (CLT)$

- $n = 25$: insufficient information for reliable calculations ($n < 30$)
- $n = 81$: $P(-0.9 < Z \leq 0.9) = \Phi(0.9) - \Phi(-0.9) = \mathbf{0.6319}$
- $n = 225$: $P(-1.5 < Z \leq 1.5) = \Phi(1.5) - \Phi(-1.5) = \mathbf{0.8664}$



> **Comment:** The probability that observations concentrate around the mean increased with the larger sample ($n = 225$).

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Example 2

Consider the independent variables:

X_i — number of points on the i-th roll of a die 🎲

Find: $P(2.5 < \bar{X} < 3.5)$ for $n = 1,\ n = 2,\ n = 36$ rolls.

Solution:

$n = 1$: $= P(X_1 = 3) = \dfrac{1}{6}$

$n = 2$:

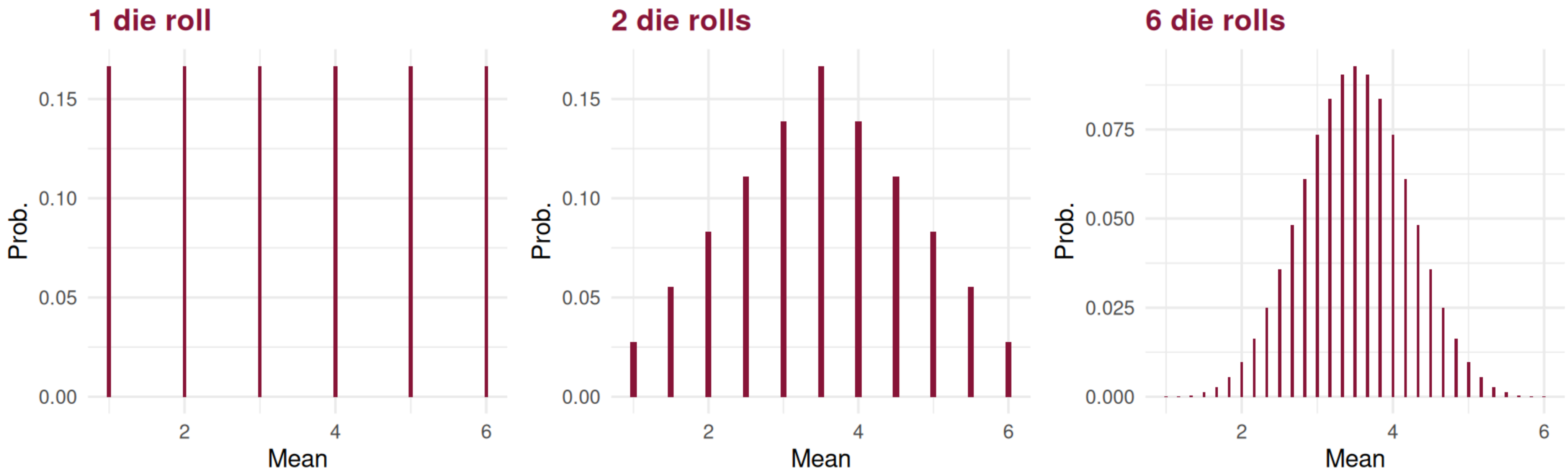
$$= P(X_1 = 1,\ X_2 = 5) + P(X_1 = 2,\ X_2 = 4) + P(X_1 = 3,\ X_2 = 3)$$
$$+P(X_1 = 4,\ X_2 = 2) + \ldots + P(X_1 = 5,\ X_2 = 1) = \frac{5}{36}$$

Example 2 — Solution (cont.)

$n = 36$ (CLT): note that for a fair die $\mu = 3.5$ and $\sigma = 1.7078$

$$= P\left(\frac{2.5 - 3.5}{\frac{1.7078}{\sqrt{36}}} < Z < \frac{3.5 - 3.5}{\frac{1.7078}{\sqrt{36}}} \right) = \Phi(0) - \Phi(-3.51) = 0.4998$$

Example 2 — Solution (cont.)



Comments:

1. The plots illustrate convergence towards the Normal distribution.
2. The variable is discrete — see the continuity correction ahead.

Corollaries of the CLT

Corollary 1

Let X be a random variable with a Binomial distribution with parameters n and p, $X \sim binomial(n, p)$. If $n \geq 30$, $np > 5$ or $n(1-p) > 5$, then:

$$X \underset{CLT}{\dot{\sim}} Normal\left(\mu = np,\ \sigma = \sqrt{npq}\right) \Leftrightarrow Z_n = \frac{X - np}{\sqrt{npq}} \underset{CLT}{\dot{\sim}} Normal(0, 1)$$

Corollary 2

Let X be a random variable with a Poisson distribution with parameter λ, $X \sim Poisson(\lambda)$. In practice, if $\lambda > 20$, then:

$$X \underset{CLT}{\dot{\sim}} Normal\left(\mu = \lambda,\ \sigma = \sqrt{\lambda}\right) \Leftrightarrow Z_n = \frac{X - \lambda}{\sqrt{\lambda}} \underset{CLT}{\dot{\sim}} Normal(0, 1)$$

Continuity Correction

> ⓘ **Note**
>
> **Note:** In both corollaries, we are approximating the distribution of a **discrete** random variable with a **continuous** (Normal) distribution. The **continuity correction** should therefore be applied:
>
> $$[F(x)]_{\text{Discrete}} \approx [F(x+0.5)]_{\text{Normal}}$$

Example: $X \sim Poisson(\lambda = 30)$

- Exact probability: $P(X = 40) = 0.01394$
- Without correction: $P\left(Z = \dfrac{40-30}{\sqrt{30}}\right) = 0$ — unusable!
- **With continuity correction:**

$$P(X = 40) = P(39.5 \le X \le 40.5) \approx P\left(\frac{39.5-30}{\sqrt{30}} \le Z \le \frac{40.5-30}{\sqrt{30}}\right)$$

$$= \Phi(1.92) - \Phi(1.73) = 0.9726 - 0.9582 = \mathbf{0.01444}$$

Continuity Correction — Other Cases

Remaining cases (discrete X, applying continuity correction):

$$P(X < 40) = P(X \leq 39) = P(X \leq 39.5) \approx \cdots$$

$$P(X > 40) = P(X \geq 41) = P(X \geq 40.5) \approx \cdots$$

$$P(35 < X < 40) = P(36 \leq X \leq 39) = P(35.5 \leq X \leq 39.5) \approx \cdots$$

$$P(35 \leq X < 40) = P(35 \leq X \leq 39) = P(34.5 \leq X \leq 39.5) \approx \cdots$$

Example 3

Consider:

1. Independent variables: $X_i \sim binomial(n_i = 1\ ;\ p = 0.05)$

   $X_i = 1$ if a tax return contains errors (otherwise $X_i = 0$)
2. Consider an inspection of a sample of $n = 1000$ tax returns.

Find the probability that there are **at least 60 incorrect** returns.

Solution: (exact and approximate distributions)

Exact distribution:

$T \sim binomial(n = 1000\ ;\ p = 0.05)$ — by the additivity of the Binomial

$$P(T \geq 60) = 1 - P(T < 60) = 1 - P(T \leq 59) = \mathbf{0.0867}$$

Example 3 — Approximate Solution

Approximate distribution (exact and approximate):

$$\mu = n_i p = 1 \times 0.05 \qquad \sigma = \sqrt{n_i p(1-p)} = \sqrt{1 \times 0.05 \times 0.95} = \sqrt{0.0475}$$

$$T \mathrel{\dot\sim} Normal(\mu_T = 1000 \times 0.05\ ;\ \sigma_T = \sqrt{1000} \times \sqrt{0.0475})$$

Applying the continuity correction:

$$P(T \geq 60) \underset{\text{cont. corr.}}{=} P(T \geq 59.5) = 1 - P(T < 59.5)$$

$$\underset{CLT}{\approx} 1 - P\left(Z \leq \frac{59.5 - 50}{6.89}\right) = 1 - \Phi(1.38) = \mathbf{0.0838}$$

> **Comment:** The variable is discrete, so we applied a continuity correction.

Example 4 [Exercise 1]

Consider: $T \sim Poisson(\lambda_T = 30)$

T = number of daily accesses to a website

Find $P(T > 40)$.

Solution: (exact and approximate distributions)

Exact distribution:

$$P(T > 40) = 1 - P(T \leq 40) = 0.0323 \simeq 0.03$$

> **Note:** Since this is a homogeneous Poisson process, the day can be viewed as a sum of n sub-periods, each with the same probability distribution.

Example 4 — Approximate Solution

Approximate distribution:

$T = X_1 + \ldots + X_n$ (by additivity), where $X_i \sim Poisson\left(\lambda = \dfrac{30}{n}\right)$

$$T \sim Poisson(\lambda = 30) \;\Rightarrow\; T \underset{CLT}{\dot{\sim}} Normal\left(\mu = \lambda = 30,\; \sigma = \sqrt{\lambda} = \sqrt{30}\right)$$

Applying the continuity correction:

$$P(T > 40) \underset{\text{cont. corr.}}{=} P(T \geq 40.5) \underset{CLT}{\approx} P\left(Z \geq \frac{40.5 - 30}{\sqrt{30}}\right) = P(Z \geq 1.92)$$

$$= 1 - \Phi(1.92) = 1 - 0.9726 = \mathbf{0.0274}$$

Approximation error: $\varepsilon = |0.0323 - 0.0274| = 0.0049$